PROBLEM scores a) 1

b) 1

c) 3

This is a **beta-alpha motif** of an enzyme family absolutely necessary for its activity

Try to answer:

a) Place the side-chains of this structure and name the model p11a.pdb TEACHER WAY:

- PDBtoSplitChain.pl -i problem1 -o problem_1_ --> will obtain the fasta seq of the motif, like the ones I had obtained
- cat problem_1_A.fa > alignment.fa
- cat problem_1_A.fa >> alignment.fa
- clustalw alignment.fa
- aconvertMod2.pl -in c -out p <alignment.aln > alignment.pir
- cp /mnt/NFS_UPF/soft/modeller/examples/automodel/model-default.py .
- modify the script and put alignment_pir, the structure is problem_1_A.pdb and in sequence = problem_1_A

Modeller doesn't just guess where atoms go; it uses Comparative Modeling (also called Homology Modeling). It follows these steps:

Reads the Alignment: It looks at your .pir file to see how your "mystery" sequence (the Target) matches up with a sequence that already has a known 3D shape (the Template).

Copies the "Skeleton": Where the sequences match, it copies the 3D coordinates from the Template's PDB file.

Satisfies Restraints: It uses a mathematical process called Spatial Restraint Satisfaction. It calculates the "ideal" distances and angles for the parts of your sequence that are different from the template.

Generates the Model: It outputs a new .pdb file containing the predicted 3D coordinates for your sequence.

```
>> Summary of successfully produced models:
Filename                          molpdf
----------------------------------------
problem_1_A.B99990001.pdb       73.39211
```

With the obtained file, we have predictes the side-chains!

b) Find the protein(s) with this motif and hypothesize what's the function of the motif. Please demonstrate your hypothesis with a sequence alignment or p-values.

BUsco el perfil .hmm i després hmmsearch sobre una seq db per trobar homòlegs

```
jackhmmer --chkhmm p11a_profile p11a.fa
/mnt/NFS_UPF/soft/databases/blastdat/uniprot_sprot.fasta
hmmsearch p11a_profile-1.hmm
/mnt/NFS_UPF/soft/databases/blastdat/pdb_seq > p11a_pdb.out
```

Even though the E-values are slightly above the standard "highly significant" threshold (0.1 instead of < 0.001), the results are very consistent, which makes them highly reliable for a short motif of only 21 residues. The top hits are all related to the Ras superfamily of small GTPases, specifically:

- RAP2A / RAP1A / Rap-1b: These are "Ras-related" proteins.
- GTPase HRas / Transforming protein P21/H-Ras: This is the classic Ras protein.

Domain annotation for each sequence (and alignments):

```
>> 1kao_A  mol:protein length:167  RAP2A
       score  bias  c-Evalue  i-Evalue hmmfrom  hmm to
alifrom  ali to    envfrom  env to     acc
 ---   ------ ----- --------- --------- ------- -------    --
----- -------    ------- -------    ----
   1 ?   13.6   0.1   0.00021      0.29       1      19 [.
5      23 ..       5      25 .. 0.90

  Alignments for each domain:
  == domain 1  score: 13.6 bits;  conditional E-value:
0.00021
```

```
  P1;p11a  1 kiamlgeggvgeaaltvnl 19
             k+ +lg ggvg++altv +
   1kao_A  5 KVVVLGSGGVGKSALTVQF 23
             7899************975 PP

>> 2rap_A  mol:protein length:167  RAP2A
       score  bias  c-Evalue  i-Evalue hmmfrom  hmm to
alifrom  ali to    envfrom  env to     acc
 ---   ------ ----- --------- --------- ------- -------    --
----- -------    ------- -------    ----
   1 ?   13.6   0.1   0.00021      0.29       1      19 [.
5      23 ..       5      25 .. 0.90

  Alignments for each domain:
  == domain 1  score: 13.6 bits;  conditional E-value:
0.00021
  P1;p11a  1 kiamlgeggvgeaaltvnl 19
             k+ +lg ggvg++altv +
   2rap_A  5 KVVVLGSGGVGKSALTVQF 23
             7899************975 PP
```

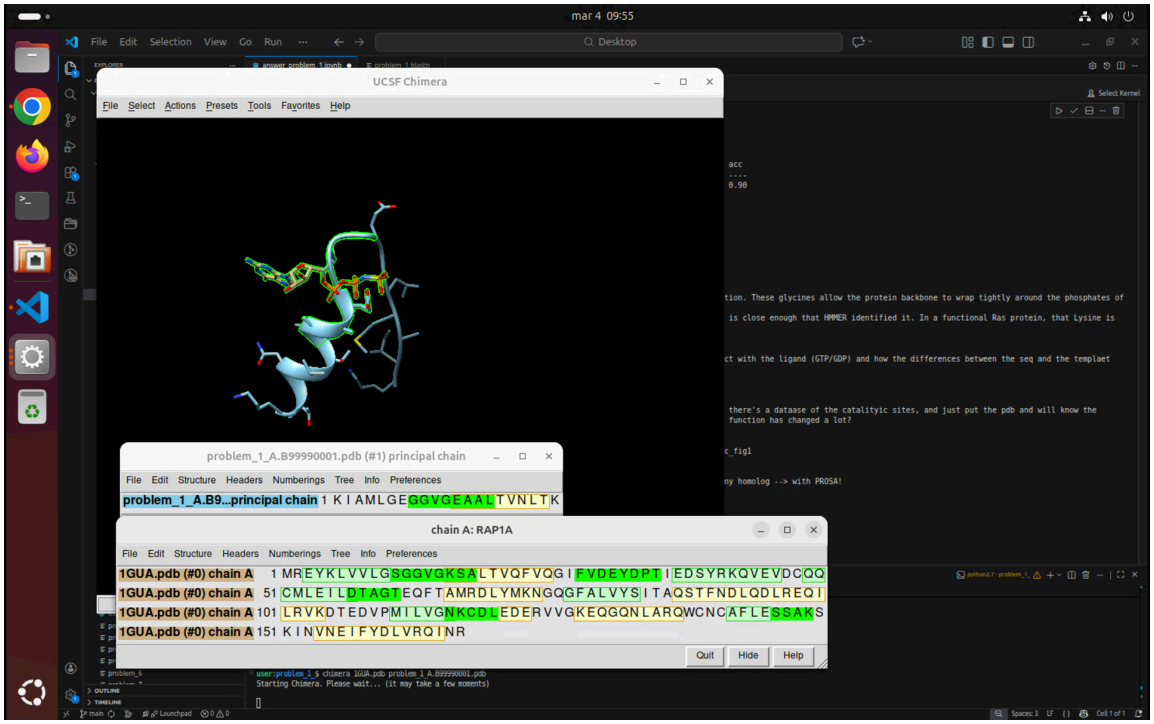
Notice the ***** underneath the GGVG stretch in the alignment. This indicates very high conservation. These glycines allow the protein backbone to wrap tightly around the phosphates of GTP. In your query, you have an E (Glutamate) where the template has a K (Lysine). However, the match is close enough that HMMER identified it. In a functional Ras protein, that Lysine is what anchors the phosphate groups

c) Do you think this motif can still be functional To determine if the motif is still functional, we must look at the specific residues that interact with the ligand (GTP/GDP) and how the differences between the seq and the templaet (RAP2A) affect that interaction.

To answer this question, show:

1. What are the Aas that do the function. Have they been mutated? To know which are the residues of the motif that interacts with the GTP molecul (theoretically a P-loop is formed GxxxxGK[S/T], where the lysine is the "anchor", as it has + charges).

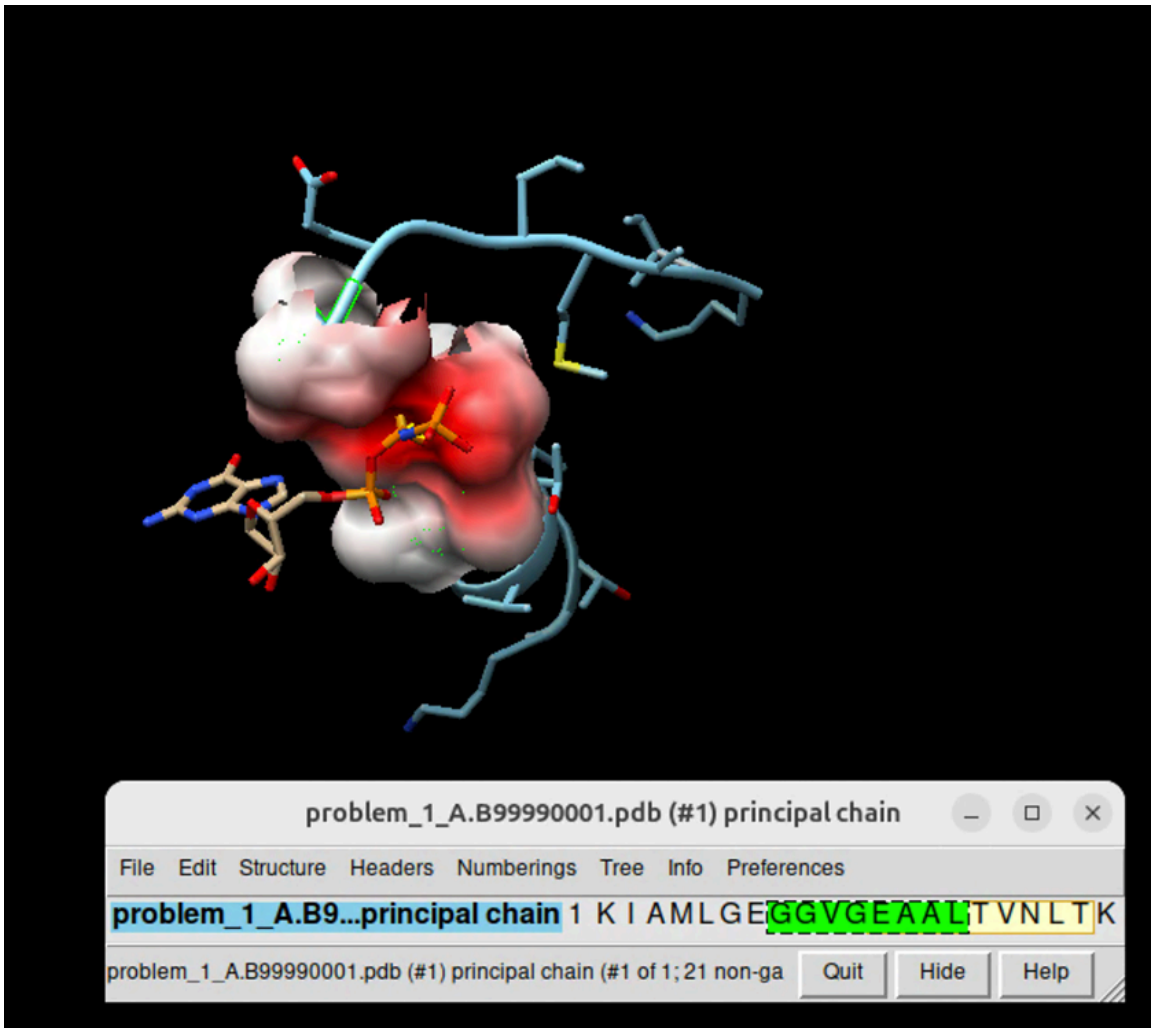
- To do so, upload the template with the GTP molecule and the predicted motif on chimera > Tools > Sequence > Sequence .
- To find nearby residues > Select > Zone > Residues



We see that in the template, there is a K (Lys) and in the motif, we have a E (Glutamate).

2. Obtain the structure of this motif bound to the ligand and name it p11c.pdb Make a JMOL/RASMOL image showing if the function can or cannot be done. Name the image p11c_fig1

- Tools > Structure Analysis > Find clashes (select the ligands and designate it, and then select the motif and also designate it, only the interaction part)
- Tools > Surface/Binding Analysis > Coulombic Surface



Lysine is positively charged and "grabs" the negative phosphates of GDP. Glutamate is negatively charged and repels them.The surface around the Glutamate (E) side chain is colored deep Red (indicating a negative electrostatic potential). The phosphate groups of the GDP molecule are also highly negative. The motif is a mutated, non-functional P-loop. While the structural scaffold (Glycines) remains, the substitution of the catalytic Lysine for Glutamate results in electrostatic repulsion with the GDP phosphates, as evidenced by the Coulombic surface coloring.

3. Calculate the statistic potential of the motif and align the profile with the profile of any homolog --> with PROSA! with known structure, as found in question "b"

d) Can you add the ATP or GDP (FER UN BLAST AMB LA SEQ DE ATP I VEUREM QUINES REGIONS INTERACTIUEN AMB AQUST P-LOOP)

blastp -query problem_1_A.fa -db /mnt/NFS_UPF/soft/databases/blastdat/pdb_seq -out problem_1.blastp gedit problem_1.blast (the e-values are really bad,)

```
chimera 1GUA.pdb problem_1_A.B99990001.pdb
```

open on chimera the superimposition and save the supeimposition with GDP in a pdb file. Unselect the whole structure, until you get the motif with the

Select the Ligand: * Go to the Select menu > Residue > and choose your ligand's name (e.g., GUA or its three-letter PDB code).

Invert the Selection:

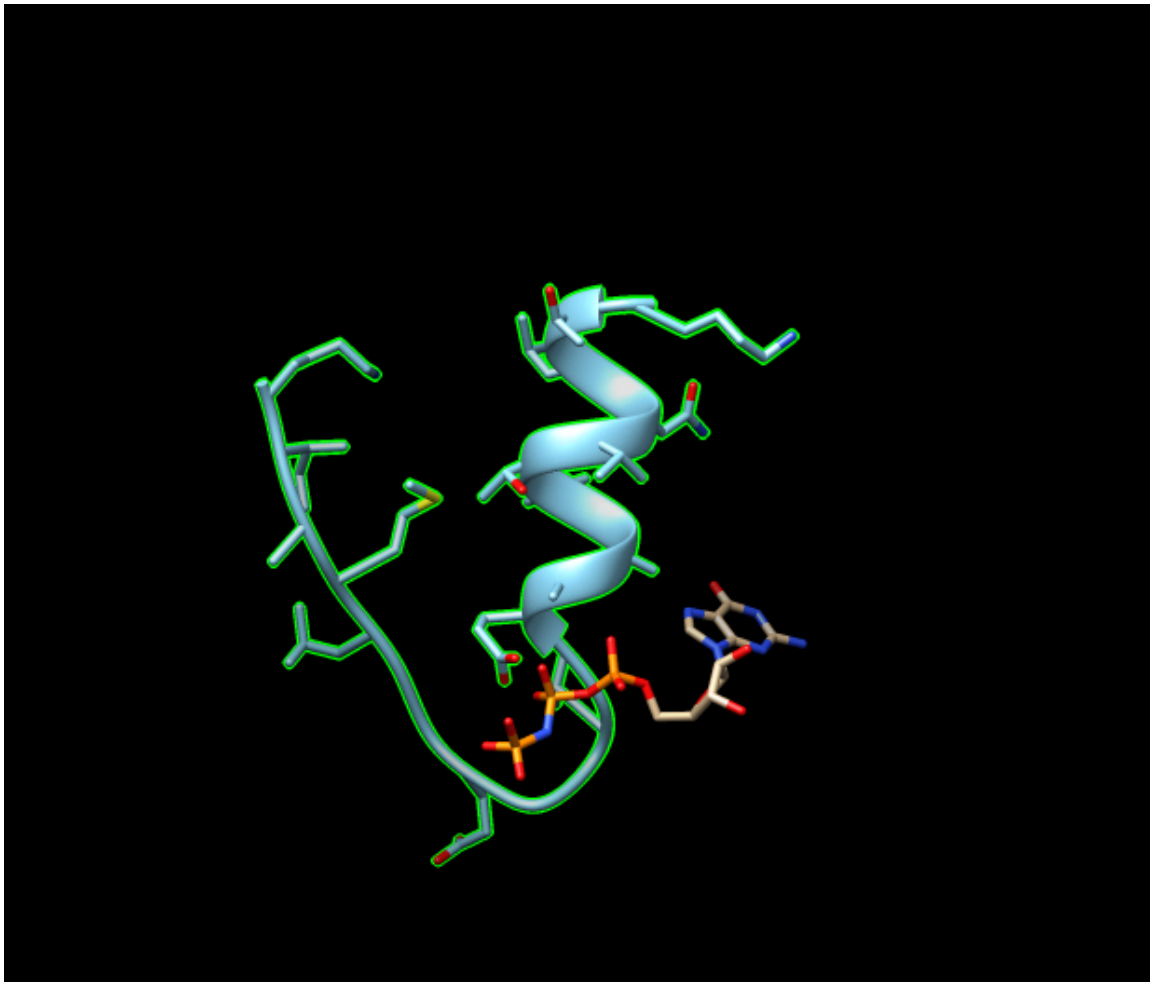
Go to the Select menu and click Invert (all models).

Now, instead of the interaction zone, everything except your zone is highlighted.

Hide the Rest:

Go to Actions > Atoms/Bonds > hide.

Go to Actions > Ribbon > hide.



MOTIF STRUCTURE

ATOM 85 CA LYS A 13 4.882 49.566 2.107 1.00 8.46 C
ATOM 107 CA ILE A 14 7.654 47.392 3.489 1.00 7.31 C

ATOM 126 CA ALA A 15 9.004 44.166 1.995 1.00 11.62 C
ATOM 136 CA MET A 16 10.650 41.347 3.995 1.00 11.38 C
ATOM 153 CA LEU A 17 13.666 39.749 2.318 1.00 7.01 C
ATOM 172 CA GLY A 18 16.039 36.995 3.438 1.00 9.05 C
ATOM 179 CA GLU A 19 16.725 33.285 3.014 1.00 9.84 C
ATOM 194 CA GLY A 20 14.306 30.682 4.311 1.00 11.26 C
ATOM 201 CA GLY A 21 14.267 30.082 8.041 1.00 11.09 C
ATOM 208 CA VAL A 22 15.792 33.403 9.159 1.00 8.09 C
ATOM 224 CA GLY A 23 12.642 34.570 10.894 1.00 10.70 C
ATOM 231 CA GLU A 24 10.804 36.735 8.336 1.00 7.87 C
ATOM 253 CA ALA A 25 7.301 35.440 9.099 1.00 9.18 C
ATOM 266 CA ALA A 26 7.962 35.295 12.858 1.00 8.07 C
ATOM 277 CA LEU A 27 9.129 38.932 12.897 1.00 6.47 C
ATOM 296 CA THR A 28 6.090 40.121 10.939 1.00 8.96 C
ATOM 310 CA VAL A 29 3.572 38.142 13.022 1.00 10.37 C
ATOM 326 CA ASN A 30 5.213 39.452 16.196 1.00 11.86 C
ATOM 340 CA LEU A 31 5.254 43.058 14.993 1.00 6.76 C
ATOM 359 CA THR A 32 1.600 43.004 13.932 1.00 11.17 C
ATOM 373 CA LYS A 33 0.318 41.320 17.107 1.00 11.12 C